

MOHIT KUMAR

Security Engineer | AI for Security & Security for AI

mohit.kr1103@gmail.com | 9973395247 | Bengaluru, India | [LinkedIn](#) | [Github](#) | [X](#) | [Portfolio](#)

EDUCATION

Vellore Institute of Technology (VIT Chennai)

Chennai, India

Bachelor of Technology in Electronics and Computer Engineering | **CGPA:** Aug 2022 – July 2026
9.12/10.0

Radiant International School

Patna, India

Higher Secondary Education | **Class XII: 95.6%** | **Class X: 94.2%**

2019 – 2021

PROFESSIONAL EXPERIENCE

ITC Infotech India Ltd.

Bengaluru, India (On-Site)

Security Engineer (IS2)

16 Jul 2026 – Present

- Promoted to full-time Security Engineer on completion of the internship, driving **AI-for-Security and Security-for-AI** initiatives — building autonomous detection tooling and hardening ML/agent supply chains alongside day-to-day SOC and VAPT operations.
- Supporting **GRC** engagements including **J-SOX** and **MICS** control auditing across IT general controls (access, change, and segregation-of-duties), testing control design and effectiveness and driving remediation of identified gaps to closure.

ITC Infotech India Ltd.

Bengaluru, India (On-Site)

Cybersecurity Intern

05 Jan 2026 – 15 Jul 2026

- Designed and deployed a multi-agent AI email security system leveraging 7 parallel ML agents, achieving 96–97% threat detection accuracy with <0.8% miss rate and 1.3% false positives, significantly improving phishing detection reliability.
- Reduced SOC email triage time from 30 minutes to a few seconds by building a real-time, explainable decision pipeline with automated threat correlation, counterfactual analysis, and SOC-ready attack narratives.
- Engineered a low-latency (<100 ms), high-throughput distributed architecture with 99.9% system uptime, enabling scalable, cost-efficient phishing detection without GPU dependency while improving analyst productivity.

PROJECTS

Bulwark — The Security Stack for Agentic AI | *Python, CycloneDX AI-BOM, SARIF, NIST AI RMF / EU AI Act*

- Built a defensive security suite for agentic-AI supply chains as three composable, deterministic scanners over one engine: **Airlock** statically disassembles model artifacts (pickle / safetensors / GGUF / ONNX / Keras) and probes live MCP servers *without ever executing them*; **Warden** imports an agent (LangChain/LangGraph, CrewAI, MCP config), builds its capability graph and scores OWASP-LLM “excessive agency” with a least-privilege before/after rewrite; **Manifest** emits a CycloneDX/SPDX AI-BOM folded into one governance report mapped to NIST AI RMF and the EU AI Act.
- On a corpus of **evasive adversarial pickle-RCE payloads**, **Airlock detected 14/14** — including the format/extension-confusion and Fickling-style import tricks that slip past picklescan, modelscan and fickling — with **zero false alarms** across benign models.
- Field-tested against **19 real public HuggingFace models**: **100%** carried at least one supply-chain finding and **95%** shipped executable pickle weights, quantifying how routinely production models ship code that runs on load.

Schema-Grounded NL→KQL — An Intermediate Representation for SIEM Rule Generation
| *LLMs, LangGraph, Pydantic, ASIM, Microsoft Sentinel*

- Borrowing intermediate-representation design from compilers, built a pipeline that turns a natural-language detection requirement into a typed, schema-validated **Security IR** grounded in Microsoft Sentinel’s ASIM schema, then *deterministically compiles* it to KQL — separating “understand the threat” from “remember the syntax,” with a closed-loop repair mechanism and an ambiguity check that asks a clarifying question instead of emitting a confidently-wrong rule.
- Evaluated two pipelines on identical inputs (direct single-prompt generation vs. IR-mediated) so every difference is attributable to the IR layer: **field validity rose from 6.7% to 86.7%** (field hallucination cut

~13×, McNemar $p < 1e-7$), syntax validity reached **97.8%**, and the repair loop recovered **96.2%** of failing cases within three attempts.

PUBLICATIONS

- **Hybrid Intensity Normalization and Translation Network for Structure-Preserving Intensity Normalization in Medical Imaging** | *Scientific Reports (Nature Portfolio)* | DOI: [10.1038/s41598-026-67051-6](https://doi.org/10.1038/s41598-026-67051-6) 2026
- **HybEx-Law: A Hybrid Neural-Symbolic System for Legal Aid Eligibility Determination** | Accepted — *Intl. Conf. on Computational Intelligence and Network Systems (CINS 2026)*, BITS Pilani Dubai Oct 2026
- **AgriCure: An AWS S3-Integrated Deep Learning Platform for Automated Crop Disease Detection** | *IEEE Xplore (i-PACT 2025)* | [Link to Paper](#) Dec 2025

TECHNICAL SKILLS

Programming: Python, Java, SQL, Prolog, Bash

Security Engineering: AI/ML & Agent Security, Software Supply-Chain Security, Kubernetes & RBAC Security, Post-Quantum Cryptography, DevSecOps & CI/CD Security, SIEM & Detection Engineering (Sentinel/KQL, Splunk), VAPT (Burp Suite, Nmap, Nessus), SOC & Threat Hunting, GRC (J-SOX, MICS), MITRE ATT&CK

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangGraph, LangChain, HuggingFace, XGBoost, OpenCV

Tools & Technologies: FastAPI, Docker, Git/GitHub, Azure AI, MongoDB, Streamlit, Jira, Jupyter

Relevant Coursework: Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms, Deep Learning, Computer Vision, NLP

CERTIFICATIONS & ACHIEVEMENTS

Certifications: [IBM Cybersecurity Analyst Professional Certificate](#) | [Stanford Machine Learning Specialization](#) | [Microsoft Azure AI Engineer Associate](#) | [Azure AI Apps and Agents Developer Associate](#) | [Azure Data Scientist Associate](#) | [University of London Cyber Security Fundamentals](#) | [OCI Generative AI Professional](#) | [Azure AI Fundamentals](#)

Achievements: IEEE IC Hackathon 2.0 Finalist (AIR 7/600 teams) | 2nd Position State-level Science Exhibition | Coordinated Bihar's first Atal Tinkering Lab (Niti Aayog funded) | Languages: English, Hindi, Bengali